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SURVEY ARTICLE

Systematic literature review on hate speech detection in Indian low-resource languages

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Abstract

Hate speech detection is a growing area of research driven by the increasing spread of harmful content on social media platforms. Automated systems are needed to identify and manage such content, particularly in low-resource Indian languages such as Hindi, Tamil, Malayalam, Telugu, and others, which remain underrepresented in Natural Language Processing (NLP) research. We conducted a Systematic Literature Review (SLR) focused on hate speech detection in these languages to assess recent developments and challenges. We retrieved 2,372 records from databases including ScienceDirect, IEEE Xplore, ACM Digital Library, Google Scholar, and arXiv, and selected 61 studies based on predefined inclusion criteria. The review highlights significant challenges such as limited annotated datasets, class imbalance, and the difficulty of modeling socio-linguistic diversity. Although multilingual and code-mixed models show promise, they often fail to capture regional language variations and emerging forms of online hate speech. Ethical concerns about bias, fairness, and transparency in automated systems are also addressed. Additionally, the influence of social norms, language variation, and platform-specific policies on model performance is discussed. This review offers valuable directions for future research, emphasizing the need for culturally informed models, better data resources, and methods to address linguistic diversity. The goal is to support the development of fair and inclusive hate speech detection systems that protect vulnerable communities while upholding freedom of expression.

Keywords Hate speech detection · Low resource language · Machine learning · Natural language processing · Offensive language detection · Social media post analysis

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Abbreviations

ADL	Anti-defamation league
BERT	Bidirectional encoder representations from transformers
CNN	Convolutional neural network
DistilBERT	Distilled bidirectional encoder representations from transformers
DL	Deep learning
EMMC	Electronic media monitoring centre
Gen Alpha	Generation alpha (born approx. 2013 onwards)
Gen Z	Generation Z (born approx. 1997–2012)
IndicBERT	Indic language bidirectional encoder representations from transformers
IPC	Indian penal code
LGBTQ+	Lesbian, gay, bisexual, transgender, queer/questioning, and others
LLM	Large language model
LSTM	Long short-term memory
mBERT	Multilingual bidirectional encoder representations from transformers
ML	Machine learning
NER	Named entity recognition
NLP	Natural language processing
PICOC	Population, intervention, comparison, outcome, and context
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RNN	Recurrent neural network
RQ	Research question
SLR	Systematic literature review
SOTA	State-of-the-art
SVM	Support vector machine
ULMFiT	Universal language model fine-tuning
XLM-RoBERTa	Cross-lingual language model – robustly optimized BERT approach

1 Introduction

Hate speech detection on social media has become more important in today's digital world because of its profound impact. Social media platforms have increased and are now a significant part of how people communicate, share ideas, and interact globally [1–3]. However, this growth has also led to a rise in hate speech, including harmful, offensive, and discriminatory content. Detecting hate speech is important to protect users, encourage respectful communication, and create a more inclusive online space. It helps prevent harm while supporting freedom of speech and the responsible use of digital platforms [4]. Addressing this issue requires thoughtful solutions that balance the right to speak freely with the need to reduce harmful content. Studying how hate speech spreads and how it can be detected is essential to make social media safer and more respectful for everyone [5]. However, it presents significant challenges due to the nature of user-generated content, which often includes low-resourced languages, slang from various generations (e.g., Gen Z, Gen Alpha), grammatical errors, typographical mistakes, and informal abbreviations.

Low-resource languages do not have enough digital data, tools, or research resources to support advanced technologies like hate speech detection. These languages are often underrepresented in research and digital platforms, making it harder to create accurate models for detecting harmful content [6, 7]. In India, many languages are considered low-resource, even though they have millions of speakers. For example, languages like Bodo, Santali, Maithili, Dogri, and Konkani have few resources for research and technology. While languages like Hindi, Bengali, Telugu, Marathi, and Tamil are more widely spoken and have more digital content, there are still gaps in resources for tasks like hate speech detection in these languages. Languages like Tamil (spoken by about 75 million people), Telugu (about 83 million speakers), and Malayalam (around 38 million speakers) face challenges in having enough data to create accurate models to detect hate speech. This makes it difficult to build reliable systems to spot harmful content in these languages [8]. Researchers are working on solutions like using transfer learning, multilingual models, and building new datasets to improve hate speech detection in low-resource languages. Developing better systems for these languages can make online spaces safer and more inclusive for people from all language communities.

A representative word cloud is shown in Fig. 1, illustrating hate speech terminologies across various Indian languages, including Tamil, Hindi, Telugu, Malayalam, Kannada, and Bengali. The visual highlights how expressions of hate vary across linguistic and cultural lines, demonstrating the need for language-specific and culturally informed approaches to automated hate speech detection. Including this figure in the

Fig. 1 Example word cloud of hate speech-related terms in multiple Indian languages

introduction helps explain the main reason for this study, which is to show the challenges of handling hate speech in low-resourced and linguistically diverse languages.

This Systematic Literature Review (SLR) has two main focuses: its scope and objectives. In terms of scope, it aims to fully explore and analyze hate speech detection on social media using advanced technologies, especially for low-resource languages. It looks at how we can better understand, evaluate, and improve the current methods in this important area, particularly for languages that do not have enough data or resources for research. As for the objectives, we want to bring together existing knowledge on this topic and create a helpful resource for researchers, practitioners, and policymakers.

Additionally, we aim to help decision-makers use hate speech detection tools responsibly, ensuring that online conversations are safe while respecting free speech. The legal and ethical dilemmas surrounding hate speech are complex, with many countries having laws against it and social media companies under pressure to moderate their platforms [9]. The presence of hate speech adversely affects user experiences, engagement, and overall trust in these platforms, making it a matter of public concern. Relevance abounds for various stakeholders. Social media users seek a safer online environment, while social media companies aim to maintain their reputation, adhere to regulations, and ensure user satisfaction [10]. Researchers and developers find opportunities for innovation, and policymakers and law enforcement agencies need practical tools to address this issue. Society benefits from addressing hate speech on social media, contributing to a more inclusive and safer online and offline environment.

Specific research objectives have been defined as follows:

1. To review how state-of-the-art algorithms deal with various types of hate speech and new online hate forms in text and multimedia, particularly in low-resource Indian languages.
2. To identify the key features of these algorithms that make them effective at detecting hate speech.
3. To explore the main challenges in using multimodal datasets for hate speech detection and how these challenges are being solved.

2 Methods

This SLR has consolidated findings from prior research on hate speech detection methods and practices. To foster a deeper understanding and advancement in this field, we have formulated ten Research Questions (RQs) based on a comprehensive review of hate speech detection strategies, their effectiveness, and real-world applications. The primary aim of these questions is to enrich our comprehension of hate speech detection and its practical implications.

RQ1. How do state-of-the-art algorithms handle different types of hate speech and new online hate forms in low-resource languages?

RQ2. What features help these algorithms effectively detect hate speech in text and other media formats?
RQ3. What are the main challenges in using multimodal datasets for hate speech detection, and how are these challenges addressed?

The databases and search options used in this SLR are highlighted in Table 1 to retrieve the publications [11]. Table 2 illustrates the Population, Intervention, Comparison, Outcome, and Context (PICOC) framework used for the SLR on hate speech detection in Indian languages, guiding the systematic analysis of publications. The framework includes Population, Intervention, Comparison, Outcome, and Context [12]. The following subsection outlines the eligibility criteria for the publications included in this SLR.

2.1 Eligibility criteria

2.1.1 Inclusion criteria

1. Types of Studies:

Table 1 Databases and search options

Database	Search options
ScienceDirect	Document type: Article Source type: Journal Keywords: Social media post analysis; hate speech; offensive language; abusive language Data range: 2017 to 2024 Language: English
IEEE	Source type: Journal Subject area: Machine learning, abusive comment analysis, offensive language detection Data range: 2017 to 2024 Language: English
ACM Digital Library	Document type: Article Source type: Journal Keywords: Social media post analysis; hate speech; offensive language; low-resource language; abusive language Data range: 2017 to 2024 Language: English
Google Scholar	Source type: Journal Subject area: Machine learning; Indian languages; offensive language detection Data range: 2017 to 2024 Language: English
Arxiv	Source type: Journal Subject area: Machine learning, hate speech detection, Indian languages Data range: 2017 to 2024 Language: English

Table 2 PICOC framework

PICOC Element	Description	Example	Keywords
Population	Research articles focused on hate speech detection in Indian languages, particularly in multilingual and code-mixed contexts, published between 2017 and 2024	Indian languages, multilingual text, code-mixed text	Hate speech, code-mixing, multilingual analysis, low-resource languages
Intervention	Various approaches and techniques, including machine learning, deep learning, multimodal analysis, and language models, designed to address challenges in detecting hate speech, such as code-mixing and multimedia content	Multimodal approaches, transformers, language models	Transformers, deep learning, multimodal analysis, LLMs, code-mixed text detection
Comparison	Comparative analysis of traditional machine learning methods versus modern approaches like transformers and large language models (LLMs). Comparisons also consider monolingual versus code-mixed language models and text-only versus multimodal approaches	Machine learning vs. LLMs, text vs. multimodal models	Traditional ML, transformers, LLMs, monolingual models, code-mixed text models, multimodal fusion
Outcome	Identification of the most effective strategies for hate speech detection in Indian languages, focusing on evolving trends to tackle challenges like code-mixing and multimodal data, improving accuracy and contextual understanding	Effective hate speech detection in multilingual contexts	Improved accuracy, contextual understanding, code-mixed detection, multimodal efficiency
Context	The research is situated in the context of hate speech detection within Indian languages, particularly emphasizing the sociolinguistic diversity and the technological challenges posed by multilingual and code-mixed environments	Social media hate speech detection in Indian languages	Indian languages, sociolinguistic diversity, multilingual datasets, code-mixing challenges

Any study design (original research) uses state-of-the-art algorithms for detecting hate speech across text or multimodal data (including text, images, audio, and video).

This includes studies evaluating methods for detecting various forms of hate speech, such as those targeting gender, LGBTQ+individuals, religious groups, political figures, children, and others.

2. Publication Date:
   Papers published from 2017 to September 2024 reflect recent algorithm advancements and hate speech detection methodologies.
3. Types of Participants/Data:
   Studies involving hate speech detection in any language, with data sources including social media, forums, news articles, and other online platforms.
4. Outcome Measures:
   Studies reporting on the performance and accuracy of hate speech detection methods, including metrics related to effectiveness across different modalities and categories of hate speech.
5. Sources:
   Peer-reviewed journal articles, conference proceedings, book chapters, preprints, blog posts, websites, and book chapters.
6. Language:

Studies published in English.

2.1.2 Exclusion Criteria

1. Types of Studies:
   Studies that do not focus on state-of-the-art algorithms for hate speech detection are not in Indian languages or are not using multimodal data.
2. Relevance:
   Studies that are not focused on hate speech detection or do not address the specific categories of hate speech in Indian low-resource languages.
3. Types of Publications:
   Review articles, opinions, editorials, preprints, blog posts, and tutorials.

2.2 Databases

Peer-reviewed Databases: ScienceDirect, IEEE Xplore, ACM Digital Library.

Grey Literature Sources: Google Scholar, arxiv.

2.2.1 Search strategy

> ((“hate speech” OR “abusive language” OR “offensive language” OR “toxic speech” OR “online abuse”)
> AND (“machine learning” OR “ML” OR “classification algorithms”)
> AND (“deep learning” OR “DL” OR “neural networks” OR “transformers”)
> AND (“large language models” OR “LLMs” OR “NLP” OR “Generative AI”)
> AND (“low-resource languages” OR “Indian languages” OR “Tamil” OR “Malayalam” OR “Telugu”)
> AND (“abusive comments” OR “toxic language” OR “online harassment”)
> AND (“women” OR “men” OR “LGBTQ+” OR “children” OR “religion” OR “politics”)).

The keywords are listed in Table 3, and the graph in Fig. 2 illustrates the number of relevant research papers per year on hate speech detection in Indian languages, retrieved from Google Scholar, spanning the years 2017 to 2024. This increase can be explained by growing concern about online hate speech, the rise of social media, and advancements in technologies like Large Language Models (LLMs), Machine Learning, and other State-of-the-Art (SOTA) techniques. These technologies, along with better resources for Indian languages, have spurred more research in this area. The peak in 2023 likely reflects the greater focus on LLMs and deep learning techniques, as well as increased funding and awareness. The drop in 2024 may suggest that the initial rush of research slowed as the focus shifted to more specialized studies and refined techniques.

Table 3 Keywords used to retrieve the articles from academic repositories

Category	Keywords
Hate speech detection types	1. Hate Speech Detection (Indian context) 2. Sexism Detection (Indian languages) 3. Cyberbullying Detection (Indian languages) 4. Racism Detection (Indian context) 5. Abusive Language Detection (Indian languages) 6. Offensive Language Detection (Indian languages)
Deep learning keywords	1. Deep Learning Hate Speech (Indian languages) 2. CNN Hate Speech (Indian languages) 3. LSTM Hate Speech (Indian languages) 4. RNN Hate Speech (Indian languages) 5. Deep Learning Cyberbullying (Indian context) 6. CNN Cyberbullying (Indian context) 7. LSTM Cyberbullying (Indian languages) 8. Deep Learning Offensive Language (Indian context) 9. CNN Offensive Language (Indian languages) 10. LSTM Offensive Language (Indian languages)
BERT-related keywords	1. BERT Hate Speech (Indian languages) 2. BERT Cyberbullying (Indian languages) 3. BERT Abusive Language (Indian context) 4. BERT Racism Detection (Indian context) 5. BERT Sexism Detection (Indian context) 6. BERT Offensive Language (Indian context)
Indian language keywords	1. Hindi 2. Bengali 3. Tamil 4. Telugu 5. Marathi 6. Urdu 7. Malayalam 8. Kannada 9.Gujarati 10. unjabi

This increase is likely driven by several factors:

1. Emerging Importance of Digital Safety: As social media and digital platforms grow in India, addressing online hate speech has become a critical concern for researchers, policymakers, and tech companies. The increase in hateful content targeting individuals and communities has driven efforts to develop automated hate speech detection systems.
2. Multilingualism and Code-Mixing: The multilingual environment in India creates unique challenges for detecting hate speech. The growing research on code-mixing uses multiple languages in the same conversation. This highlights the complexity of Indian languages in hate speech detection.
3. Multimodal Analysis: Another growing trend is the application of multimodal approaches in hate speech detection. This review identifies that research in this area shows an increasing use of combining text with other media (such as images and audio) to improve detection accuracy. With social media becoming

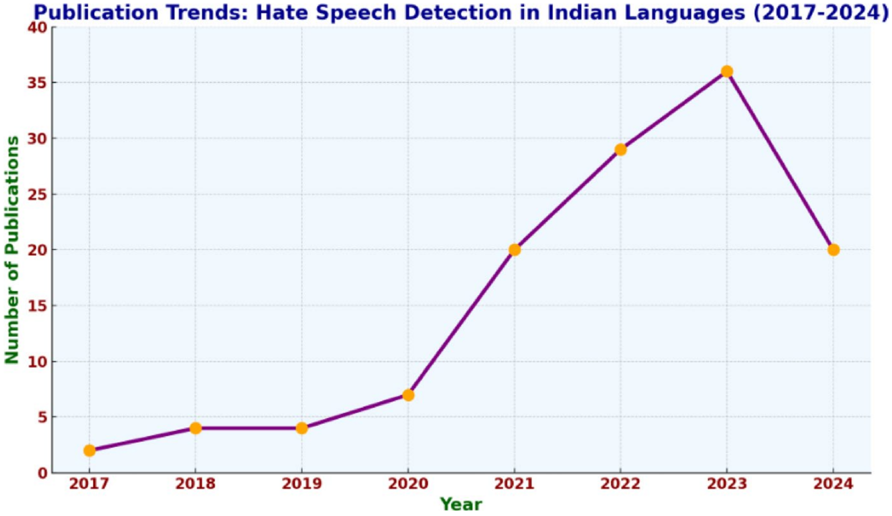


Fig. 2 Year-wise analysis of research publications on hate speech detection in Indian languages (2017–2024)

more visually-centric, leveraging multimodal data helps capture context more effectively.

4. Technological Advances: Advances in natural language processing (NLP), Machine Learning, and Deep Learning have likely contributed to this rapid increase in research. The availability of more sophisticated models, including transformers and large language models, has opened up new possibilities for tackling the linguistic and contextual challenges inherent in hate speech detection, particularly in low-resource languages such as Tamil, Malayalam, and Telugu.

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the Systematic Literature Review (SLR) is presented in Fig. 3 [13–15]. The graph in Fig. 4 summarizes all the papers considered for this study in the span of 2017–2024 and provides an overview of the papers collected from various journals and conferences. Each journal and conference has contributed to the growing research on hate speech detection, particularly in multilingual and low-resource contexts. Journal-wise Breakdown: The articles in this study come from diverse sources, including IEEE Transactions Journals, Springer, Elsevier, ACM, Cold Spring Harbor Laboratory Press, and others. These journals are recognized for publishing high-impact research in natural language processing, machine learning, and computational linguistics.

2.2.2 Year-wise trend

From 2017 to 2020, a steady but modest number of papers were published, with contributions primarily coming from ACM Transactions, Elsevier, and IEEE. The research focus was still emerging during this period, with early-stage methodolo-

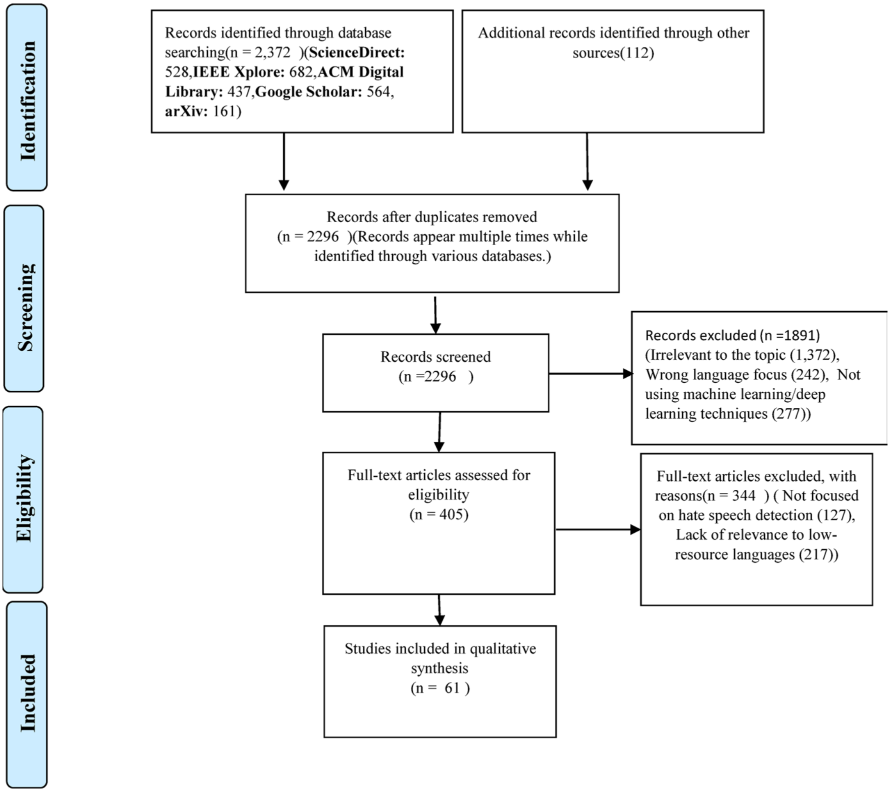


Fig. 3 PRISMA flow diagram of SLR

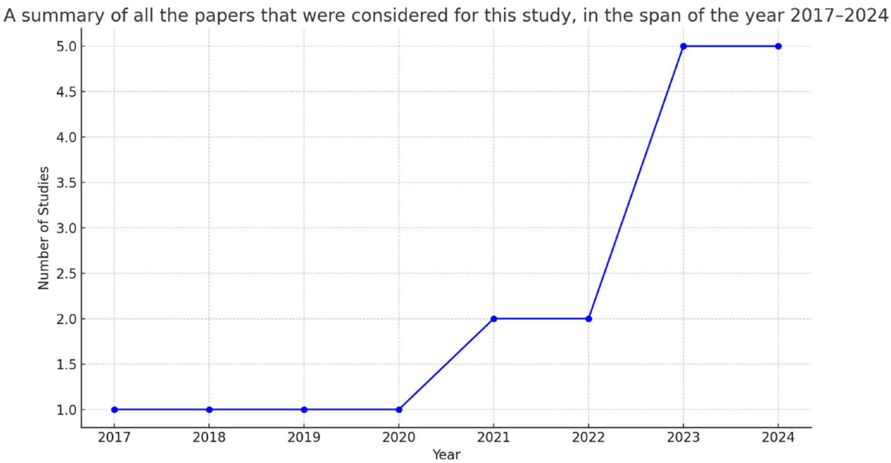


Fig. 4 Year-wise distribution of collected papers (2017–2024), showing the increasing trend in the number of studies considered for this review over time

gies in detecting hate speech. In 2021 and 2022, the number of publications saw a significant rise. This period reflects the impact of technological advancements, such as transformer-based models, allowing for more effective detection methods. IEEE, Springer, and Elsevier contributed significantly to the growth of research output during this period.

By 2023 and 2024, the contributions peaked, particularly from IEEE Conferences, Springer Journals, and Elsevier. These journals have published many studies focused on innovative approaches to handling complex issues such as code-mixing, multimodal hate speech detection, and processing low-resource languages.

2.2.3 Key insights by publisher

1. IEEE Transactions and Conferences: IEEE contributes significantly with high-quality conference proceedings and journal papers. These papers primarily focus on machine learning techniques, transformer-based models, and hate speech detection methods designed for Indian and low-resource languages.
2. Springer: Springer Journals have consistently increased publication volume, particularly in 2022 and 2023. These papers often explore advanced Deep Learning techniques, data augmentation methods, and domain-specific approaches to multilingual hate speech detection.
3. Elsevier and ACM: Both publishers have contributed consistently, providing research papers that apply novel computational techniques to hate speech detection. Elsevier publications focus heavily on social media applications and the ethical implications of hate speech, while ACM emphasizes computational models and code-mixed language processing.

3 Research growth

The overall trend across different publishers shows an apparent increase in research interest and output from 2017 to 2024. This reflects the growing importance of handling hate speech in multilingual and multimedia-rich environments, especially in the context of Indian languages. The consistent growth from 2021 onwards also indicates the maturing of this research field, supported by the rapid development of NLP technologies. The chord graphs below are derived from the data presented in Table 4, highlighting the distribution and focus areas of the related literature.

The collaboration patterns depicted in Fig. 5 reflect the growing international interest in addressing hate speech in Indian low-resourced languages. India plays a central role, often partnering with countries like Ireland, Denmark, the United States, and Australia in multilingual studies. These collaborations are essential for developing context-aware models and for generating annotated datasets that support underrepresented languages such as Tamil, Malayalam, Kannada, and Marathi. The inclusion of multiple linguistic and regional perspectives strengthens the scope of research and highlights the need for sustained global cooperation in tackling culturally nuanced online abuse.

Table 4 Descriptions of related works

Author	Country	Cited by	Language	Code mixing	Method	Dataset	Weighted F1	Accuracy
Bohra et al. [32]	India	295	Hindi + English	✓	SVM	Tweets	–	0.717
Roy et al. [8]	India	79	Tamil + Malayalam	✓	A deep ensemble learning framework	Tweets + YouTube comments	0.933	–
Rani et al. [61]	Ireland	61	Hindi + English	✓	character-level CNN	Facebook + Tweets	0.74	0.86
Ghosh et al. [62]	India	19	Hindi, Marathi, and Bengali	–	Transformer-based Language Models	HASOC-Hindi (2019), HASOC-Marathi (2021), and Bengali Hate Speech (BenHateSpeech)	0.909	0.909
Nandi et al. [63]	India	2	Bengali, Hindi, and Marathi	–	stacked ensemble of fine-tuned mBERT and indicBERT	BD-SHS, HASOC, and L3Cube-MahaHate	0.923, 0.815, and 0.924	0.923,0.814, and, 0.923
Sreelakshmi et al. [64]	India and Ireland	1	Kannada-English, Malayalam-English, and Tamil-English	✓	Transformer embedding with Machine Learning classifiers	HASOC 2020 Malayalam, HASOC 2021 Malayalam, HASOC 2021 Tamil, DravidianLangTech 2021 Malayalam, DravidianLangTech 2021 Tamil, and DravidianLangTech 2021 Kannada	–	–
Ghosh et al. [6]	India	–	Hindi, Marathi, and Bangla	–	–	HASOC-Hindi, HASOC-Marathi, and HS-Bangla	–	–
Ghosh et al. [65]	India and Ireland	89	Malayalam, Tamil, and Kannada	✓	ULMFit and DistilmBERT	DravidianLangTech-EACL2021	0.9603, 0.7895, and 0.727	

Table 4 (continued)

Author	Country	Cited by	Language	Code mixing	Method	Dataset	Weighted F1	Accuracy
Farooqi et al. [66]	India	40	Hindi + English	✓	a hard voting ensemble of IndicBERT, XLM-RoBERTa, and Multilingual BERT	HASOC 2021	0.72	–
Ghosh et al. [67]	India	12	Hindi + English, and Marathi	✓	transfer-based BERT model	HASOC 2022	0.66	–
Jafri et al. [68]	India, United States, Ireland, Australia	1	Hindi	–	Hard Ensemble of BERT-Models	Tweets	0.733	0.935
Bhatnagar et al. [69]	India	12	Hindi	–	Pre-trained transformer encoders	provided by the competition organizers	0.93	–
Pathak et al. [70]	India, Denmark	34	Malayalam, Tamil	–	Custom word embedding	HASOC organizers	0.77,0.87	–

The interconnections among languages used in hate speech detection studies related to the Indian low-resource setting are highlighted in Fig. 6. Hindi and English show the highest frequency of co-occurrence, indicating their dominance in bilingual and code-mixed datasets. Tamil, Malayalam, and Kannada also exhibit strong interconnection in studies targeting South Indian languages. The presence of languages such as Bengali, Marathi, and Telugu in multiple pairings reflects growing attention toward regional diversity. These language-level collaborations demonstrate an increasing effort to build inclusive models that can address the multilingual realities of hate speech across India.

It reveals that methods like transformer-based models, ensemble learning approaches, and BERT variants are widely used across multiple Indian languages, including Hindi, Tamil, Malayalam, Marathi, and Kannada, as illustrated in Fig. 7. Notably, Hindi and English appear most frequently, often in code-mixed or bilingual datasets. The diagram also highlights that transformer models are increasingly used for Indian languages, suggesting a growing focus on low-resource settings.

Hindi (30% e.g., HASOC, HOT), Tamil (20% e.g., DravidianLangTech, Tamil Offensive Speech), Malayalam (15% e.g., DravidianLangTech-Malayalam), Marathi (10% e.g., Marathi Hate Speech Corpus), Bengali (10% e.g., HASOC-Bengali),

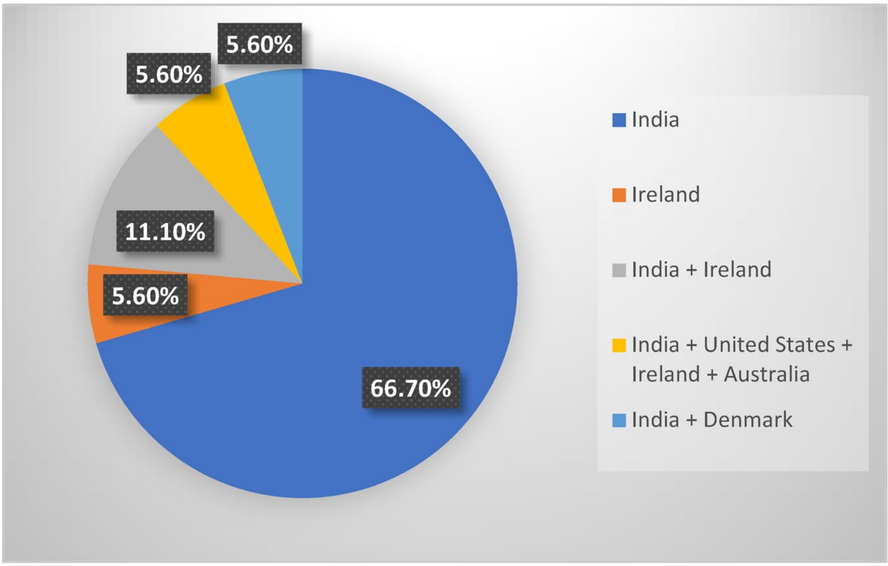


Fig. 5 Country-level collaborations in hate speech detection studies. *Note*: Pie chart illustrating country-level collaboration types in hate speech detection studies on Indian low-resource languages. India dominates single-country studies, with more minor but significant collaborations involving Ireland, Denmark, the United States, and Australia

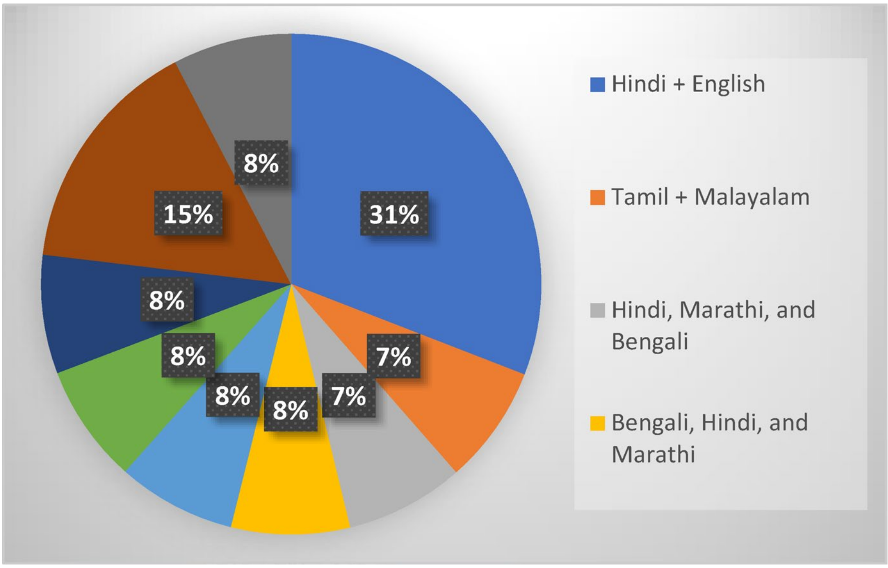


Fig. 6 Language-level collaborations in hate speech detection research. *Note*: Pie chart showing the co-occurrence of Indian and associated languages used in annotated datasets and model training for hate speech detection. Each arc represents a language, and the links reflect the frequency of language pair usage across reviewed studies

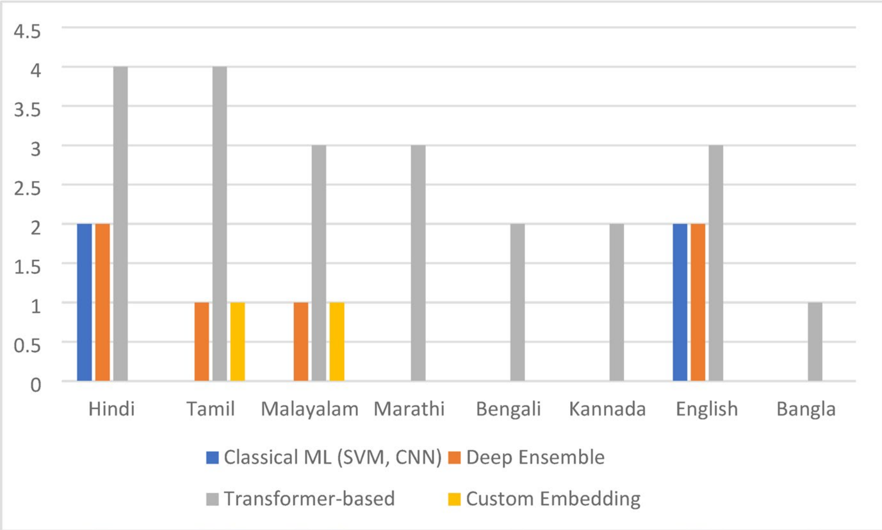


Fig. 7 Method–language mapping in hate speech detection research. *Note*: It illustrates the association between various machine learning approaches and the languages they have been applied in hate speech detection studies. Each arc represents either a method or a language, and the connecting lines indicate their usage in combination. The number of connecting strands visually represents the frequency of method-language pairings

English (10% e.g., HASOC-English), and Kannada (5% e.g., DravidianLangTech-Kannada). The varying levels of dataset availability for different Indian languages in hate speech detection research are illustrated in Fig. 8. Datasets such as HASOC and DravidianLangTech, created by shared task organizers, demonstrate structured multilingual support, often including Hindi, English, Tamil, and Malayalam. In contrast, datasets like Tweets, Facebook + Tweets, and YouTube comments were curated independently by researchers using data scraped directly from social media platforms. These user-created datasets tend to reflect real-world usage patterns and code-mixed expressions. It highlights how both curated competition datasets and organically constructed corpora contribute to supporting multiple Indian languages, indicating the field's ongoing efforts to balance standardization with linguistic diversity.

4 Background

Hate speech refers to verbal, written, or symbolic communication that expresses prejudice, hostility, or violence towards individuals or groups based on attributes such as their ethnicity, religion, gender, sexual orientation, disability, or other characteristics. Hate speech aims to degrade, discriminate against, or incite harm or violence against the targeted individuals or groups [4, 16, 17].

Hate speech can take various forms, as illustrated in Table 5. It is essential to note that while free speech is a fundamental right, hate speech is typically considered harmful and outside the bounds of protected speech in many legal and social contexts

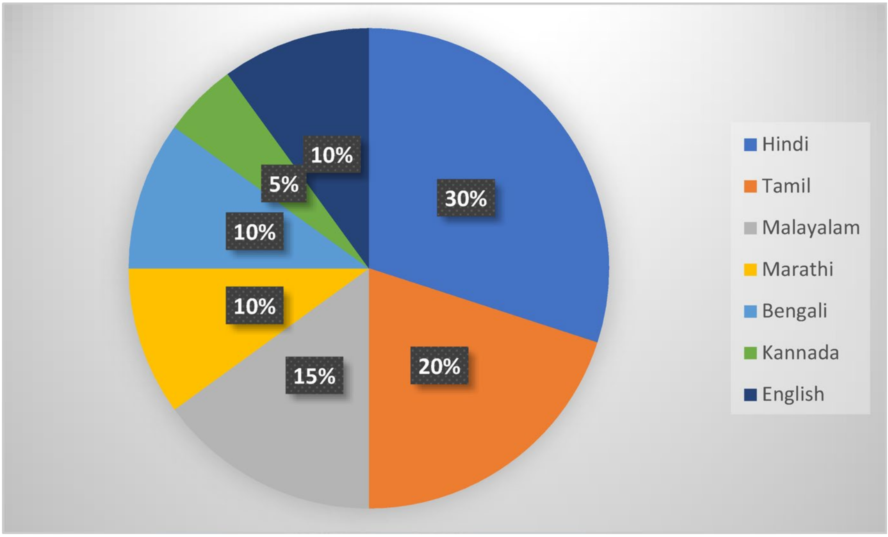


Fig. 8 Language distribution across datasets in hate speech detection. *Note*: Pie chart showing the distribution of languages across datasets in hate speech detection research involving Indian low-resource languages. Each slice represents a language, and its size indicates the proportion of datasets covering that language

due to its potential to incite discrimination, violence, or harm to individuals or marginalized groups. Laws and regulations concerning hate speech vary by jurisdiction [10, 18–21].

The visualization of the multifaceted dimensions of hate speech described in Table 5 is depicted in Fig. 9. The prevalence and impact of Hate speech on social media involve many challenges. Hate speech is widely disseminated across various social media platforms, with users taking advantage of these platforms' speed and anonymity. Its psychological impact is profound, causing emotional distress, anxiety, and insecurity among targeted individuals [22, 23]. Moreover, it continues to promote discrimination and prejudice based on attributes such as race, religion, gender, and sexual orientation. Hate speech can escalate into cyberbullying and harassment, further harming victims. Marginalized communities are particularly vulnerable, as hate speech exacerbates existing inequalities and isolation. The fear of encountering hate speech can silence individuals, impeding free expression and open dialogue.

Furthermore, it has real-world consequences, from inciting violence to damaging the reputation of social media platforms and eroding user trust [24]. Legal and regulatory implications add to the complexity, placing social media companies under pressure to moderate content while respecting freedom of expression [25, 26]. Grasping the full extent of these issues is essential for developing effective strategies to mitigate the harm of hate speech and strike a balance between protecting individuals and preserving free speech online.

Figure 10 presents monthly reports in billions, detailing the active users of globally popular social media platforms. Some relevant statistics and case studies highlight the prevalence and impact of hate speech on social media as follows:

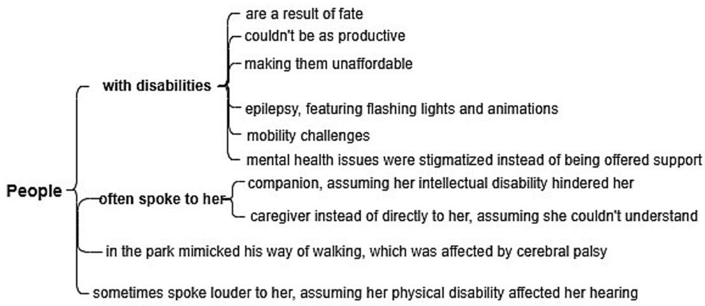
Table 5 Various forms of hate speech

Types of hate speech	Description
Ethnic Hate Speech	This involves derogatory comments, slurs, or offensive content aimed at people of a specific racial or ethnic background. It often perpetuates stereotypes and fosters racial discrimination [71]
Religious Hate Speech	This form targets individuals or groups based on their religious beliefs, often involving offensive remarks, threats, or derogatory content aimed at a particular faith or its followers [17, 18]
Homophobic and Transphobic Hate Speech	Targeting individuals based on their sexual orientation or gender identity. It includes slurs, derogatory comments, or calls for violence against LGBTQ+individuals [19, 20, 72]
Sexist and Misogynistic Hate Speech	This form denigrates individuals based on their gender, often perpetuating stereotypes, objectifying, or demeaning them [21, 43, 73–75]
Ableism	Hate speech is directed at individuals with disabilities, often involving derogatory comments or jokes that demean or devalue their capabilities [18, 19]
Xenophobic Hate Speech	Targeting individuals or groups based on their national origin or immigration status involves derogatory remarks, stereotyping, or calls for exclusion [21]
Political Hate Speech	Expressing hostility, threats, or derogatory language towards individuals or groups with differing political beliefs or affiliations [22, 76, 77]

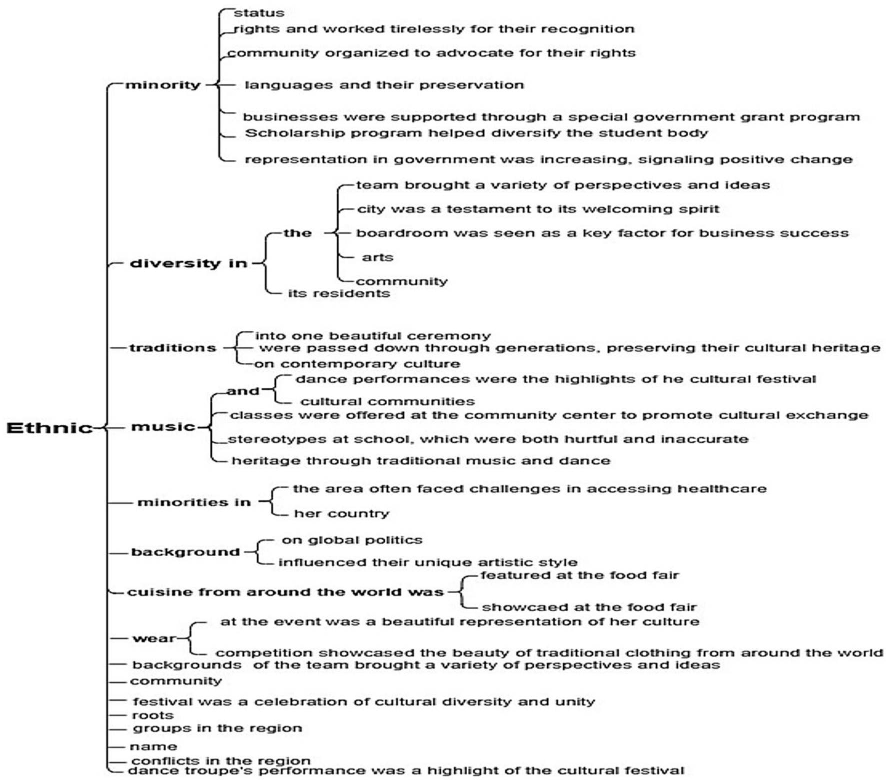
This table outlines key categories of hate speech based on targeted identity attributes such as ethnicity, religion, gender, sexual orientation, disability, and political affiliation. Understanding these distinctions is essential for developing context-aware hate speech detection systems and for addressing the diverse forms of harm experienced across online platforms.

This digital landscape has seven social media platforms, each boasting over one billion monthly active users. Remarkably, four of these seven platforms are under the ownership of Meta. Additionally, 15 social media platforms can claim to have at least 400 million active users during this period. Facebook stands out with an impressive 3.030 billion monthly active users, while YouTube follows closely with a potential advertising reach of 2.491 billion. As of July 2023, Facebook continues to maintain its position as the world's most widely used social media platform.

WhatsApp also stands strong, with a user base exceeding 2 billion. Instagram has 2 billion monthly active users, and WeChat, including its Chinese counterpart Weixin, boasts 1.327 billion monthly active users. TikTok has the potential to reach a substantial audience of 1.218 billion adults over 18 each month, and Facebook Messenger offers an advertising reach of 1.036 billion. Telegram claims 800 million monthly active users, while Snapchat has 750 million monthly active users. Douyin has successfully gathered 743 million monthly active users while enjoying 673 million. Twitter's reported potential advertising reach for July 2023 was 666 million, though it is important to note that the platform's published ad reach data is subject to significant fluctuations, even daily. Sina Weibo boasts 599 million monthly active users, QQ has 571 million, and Pinterest completes the list with 465 million monthly active users [27]. This data underscores global social media platforms' vast and diverse landscape.

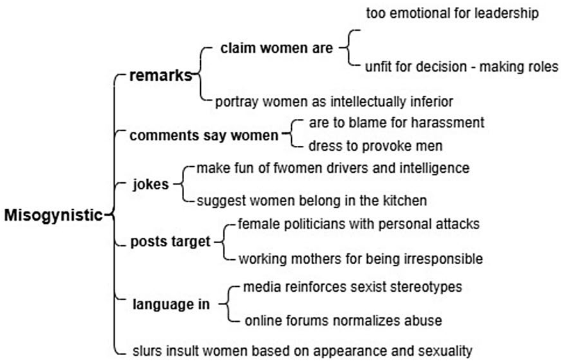


(a). Word tree for examples related to Ableism.

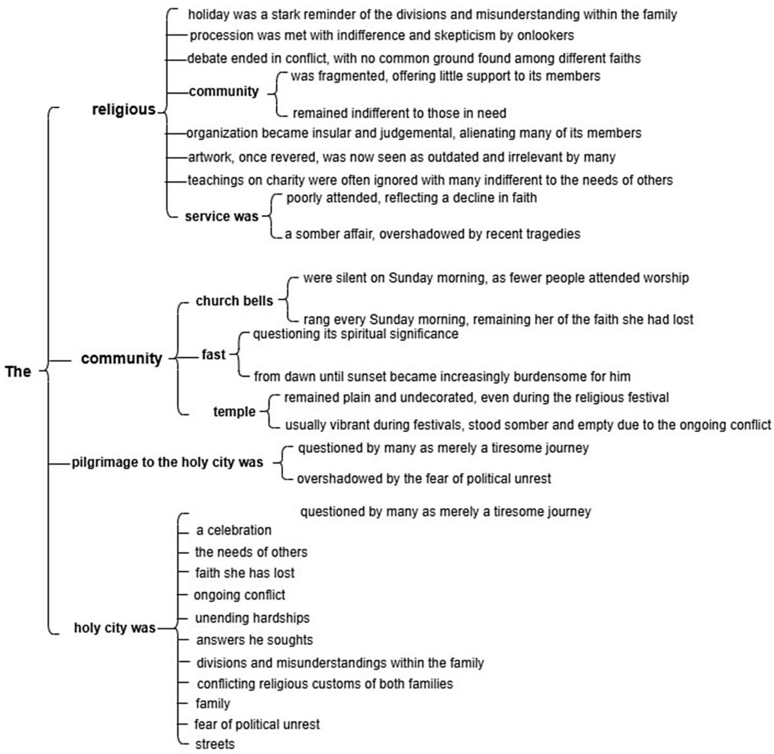


(b): Word tree for examples on Ethnic

Fig. 9 Word tree representation of various forms of Hate Speech (**a** Ableism, **b** Ethnic, **c** Misogyny, **d** Religious, **e** Political, **f** Homophobic and Transphobic Hate Speech, **g** Xenophobic hate speech). *Note*: In the visualizations presented, word tree diagrams were used to explore linguistic patterns associated with hate speech directed at marginalized communities. Each diagram begins with a central root word, such as *People*, *LGBTQ+*, *Misogynistic*, or *Xenophobic,* which serves as the anchor point from which all associated phrases extend. These root terms represent the primary subject or group being targeted, and the branching structures illustrate how language evolves around them to convey harmful or discriminatory messages. By centering the analysis on these root expressions, the visualizations enhance clarity and guide the reader's attention to the core linguistic constructions that surround hate speech across different contexts



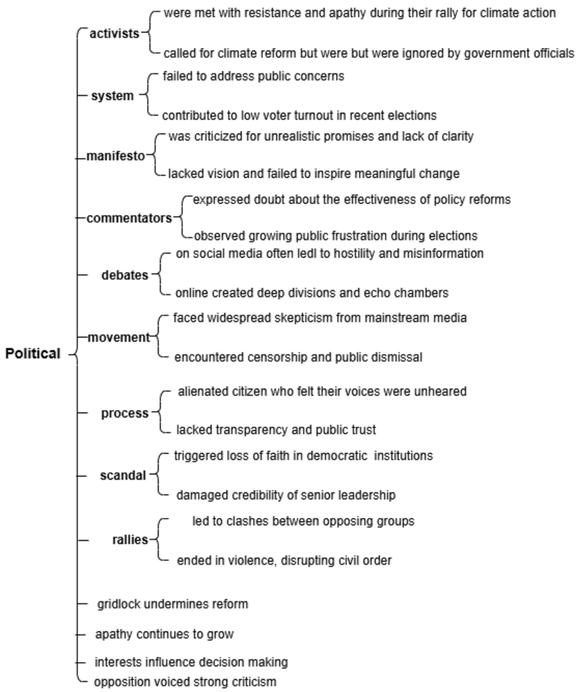
(c) Word tree for example on Misogyny



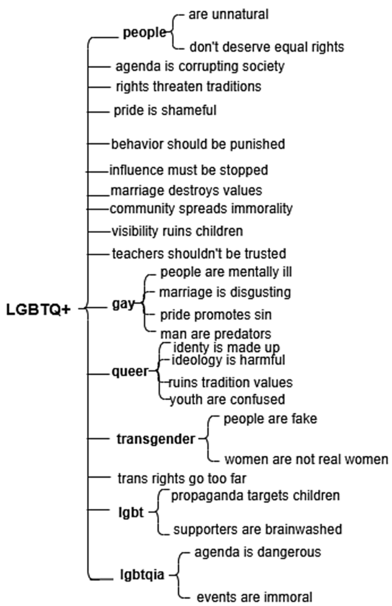
(d) Word tree for example on Religious Hate Speech

Fig. 9 (continued)

Table 6 illustrates a few case studies examining instances of hate speech in different locations worldwide. In 2020, the Anti-Defamation League (ADL) reported a disturbing trend of more than 4.2 million tweets containing anti-Semitic language (prejudice, hostility, or discrimination against Jewish people), shedding light on the alarming increase in hate speech targeting religious groups [28]. Moreover, a study

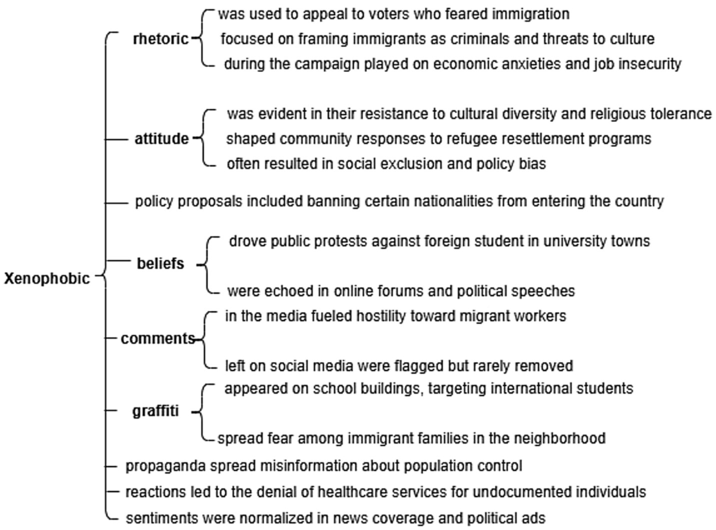


(e) Word tree for example on Political Hate Speech



(f) Word tree for example on Homophobic and Transphobic Hate Speech

Fig. 9 (continued)



(g) Word tree for example on Xenophobic hate speech

Fig. 9 (continued)

by the Pew Research Center found that 59% of US teens have encountered some form of online harassment or hate speech, emphasizing the vulnerability of young internet users to such abuse (Reporting – CYBER BAAP, n.d.). Gender-based hate speech is another pervasive issue, with a report from the United Nations Broadband Commission revealing that 73% of women have experienced online violence, including hate speech and threats [29].

The Gamergate controversy began within the online gaming community in August 2014 and has spread across multiple social media platforms. There have been widespread incidents of cyberbullying and online aggression. The author [80] examined this Gamergate controversy by curating a dataset with 340,000 unique users and 1.6 million tweets to understand the behaviour of the users and the nature of their content, and how they differ from a baseline of random Twitter users. The findings revealed that individuals involved in this “Twitter war” generally have more friends and followers, display higher levels of engagement, and produce negative sentiment and increased hate-related content compared to random users. In addition, they conducted preliminary evaluations of how Twitter’s suspension mechanisms address such abusive behaviors. Lastly, political polarization has been recognized as a contributing factor, as a Pew Research Center study indicates that 64% of Americans believe fake news and disinformation play a role in fostering political polarization, creating an environment where hate speech can flourish [30].

Table 7 summarizes prominent initiatives, including Hatefusion, developed by IIIT-Allahabad, which classifies social media content into hate, offensive, and non-hate categories [31]. Similarly, startups like Samvidhaan.AI and platforms such as Tattle Civic Tech focus on real-time monitoring and multilingual support, particularly on platforms like WhatsApp and Twitter. Academic and collaborative projects involving IIT Madras, Wipro AI Lab, and MIT CSAIL are also contributing signifi-

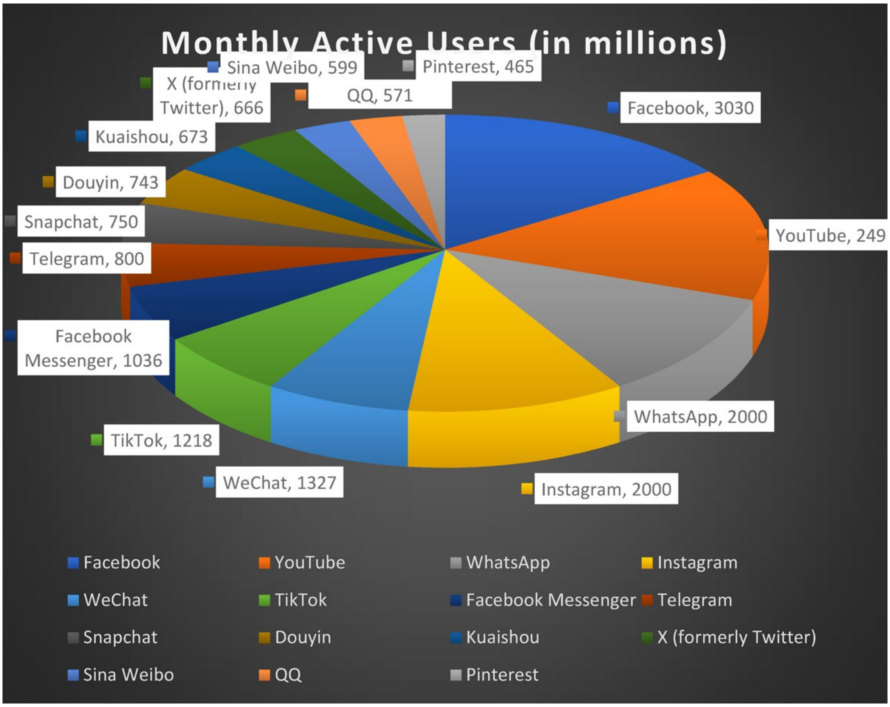


Fig. 10 Worldwide popular social media platforms. *Note*: The above figure presents the monthly active users (in billions) of the most popular global social media platforms, based on data accessed from Statista in November 2023 [27]. Since Statista regularly updates its content, an archived version of the source has been preserved for verification and reproducibility. Archived link: https://web.archive.org/web/20231203095547/https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/

cantly by creating datasets and detection models targeting subtle and coded forms of hate in Indian languages. These tools collectively represent a growing ecosystem aimed at curbing online toxicity and fostering digital accountability in India [32, 33].

We compared temporal search trends using Google Trends to validate the contextual significance of selected case studies. As shown in Fig. 11, spikes in search interest often align with major hate speech incidents. For instance, in the Tablighi Jamaat congregation in Delhi in 2020, a surge of Islamophobic content appeared on Twitter. Hashtags such as #CoronaJihad falsely accused Muslims of deliberately spreading COVID-19, leading to widespread stigma, misinformation, and increased communal tension during the pandemic. The "Sulli Deals" controversy (July 2021), in which Muslim women were targeted via a fake auction app, coincided with a sharp increase in search volume for hate speech-related terms. Similarly, the "Bulli Bai" incident in January 2022 and various communal flare-ups during political events led to recurrent spikes. These correlations suggest that such events not only provoke hate content online but also stimulate broader public discourse, reinforcing their relevance for inclusion in this review.

Table 6 Case studies

Author	Case studies	Description
Backe et al., [78], Massanari [79], Chatzakou et al., [80]	Gamergate	The Gamergate controversy in 2014 highlighted how hate speech on social media can escalate into harassment campaigns. Female gamers and game developers faced severe online abuse, including threats of violence and doxing
Algül and Akpınar [24], Kozyreva et al., [25], Lévesque [26],Hellmann [81]	Facebook's Content Moderation Challenges	Facebook has faced numerous challenges in moderating hate speech on its platform. The platform has struggled to balance free expression and content removal. The company's transparency reports showcase the sheer volume of content that requires moderation
Bromell [82],Bromell [83], Calhoun and Weston [84],Arda [85]	Christchurch Mosque Shootings	The perpetrator of the 2019 Christchurch mosque shootings in New Zealand live-streamed the attack on Facebook, revealing the platform's vulnerability to the spread of hate speech and violence
Gillespie [86], Jhaver et al., [87], Davidson et al., [88], Matias [89], Konikoff [90],	Twitter and Harassment	Various high-profile individuals, particularly women and minorities, have experienced extensive hate speech and harassment on Twitter. Cases like these have raised questions about the platform's ability to protect users
Chandra et al., [91], Tieri and Ranjan [92], Ghasiya and Sasahara [93]	Islamophobic Hashtags During COVID-19 ("Corona Jihad")	In 2020, the Tablighi Jamaat congregation in Delhi, a flood of Islamophobic content emerged on Twitter, with hashtags like #CoronaJihad accusing Muslims of spreading COVID-19 intentionally. This caused widespread stigma and communal tension during the pandemic
Jagani [94]	Targeted Harassment of Muslim Women via "Sulli Deals" & "Bulli Bai" Apps (2021–2022)	Two GitHub-hosted apps were created to "auction" Muslim women without their consent, using publicly available photos. This coordinated misogynistic and Islamophobic attack sparked national outrage and highlighted the vulnerability of women from minority communities online

This table summarizes key real-world case studies that demonstrate the complexity and societal impact of hate speech on online platforms. Each case highlights different challenges, including platform moderation limitations, targeted harassment, gendered and religious bias, and the spread of harmful content during crises. These incidents underline the urgency of developing robust, context-aware hate speech detection systems.

5 Methods of hate speech detection

Hate speech detection is a critical area of research and technology development aimed at identifying and mitigating offensive, discriminatory, and harmful content on the Internet and other digital platforms. Various approaches and techniques are used in hate speech detection, and they can be categorized into several broad categories.

Table 8 illustrates different techniques along with their descriptions and associated limitations. The evolving landscape of hate speech regulation on social media features a series of methodological advancements from various research fields. Researchers [34, 35] have pioneered Lexical and text-based methods that rely on identifying offensive words, n-gram patterns, and sentiment cues. While these approaches are easy to implement, they often fail to understand the context, leading to false positives. They are also ineffective against new or disguised forms of hate speech and struggle to generalize across languages and platforms. Machine learning and natural language processing (NLP) methods offer more flexibility by learning from data using supervised or unsupervised models and deep neural networks. However,

Table 7 Hate speech detection tools and initiatives in India

Tool/project name	Developed by	Description	Key features / notes
Hatefusion	IIIT-Allahabad	Detects hate speech in social media posts	Analyzed 24,783 posts; identified 5.77% as hate speech, 16.8% as non-hate, the rest as offensive
Samvidhaan.AI	Private Startup (with public interest focus)	AI-powered tool to track hate speech and misinformation	Focuses on multilingual detection, especially during elections and communal tension periods
Tattle Civic Tech	Independent initiative	Detects harmful and misleading content on WhatsApp	Works via crowd-sourcing messages and applying machine learning-based classification
INDIA@75 Hate Speech Dataset	Wipro AI Lab & IIIT Hyderabad	Dataset, not a tool, but foundational for tool development	Helps build hate speech models for Indian languages; includes annotations
MIT-CSAIL+IIT Madras Collaboration	MIT (USA) & IITM	Research to detect coded hate speech in Tamil and Hindi	Uses multilingual models and contextual embeddings
Hate Speech Classifier—CDAC	Centre for Development of Advanced Computing	A tool for classifying toxic and hate speech	Focus on the Indian government and defense communication screening

This table highlights key Indian efforts in detecting hate speech through technological tools, research collaborations, and civic tech platforms. The tools vary in their approach, language coverage, and application domains, ranging from academic prototypes to real-time civic interventions.

these models are highly dependent on data quality and are often difficult to interpret. They face challenges in understanding sarcasm, implied hate, and culturally sensitive expressions. Moreover, these systems require significant computational resources and are difficult to deploy across multilingual and evolving content environments. Machine Learning and Natural Language Processing (NLP) [4, 36] have made significant strides, enabling computers to learn from data for enhanced language under-

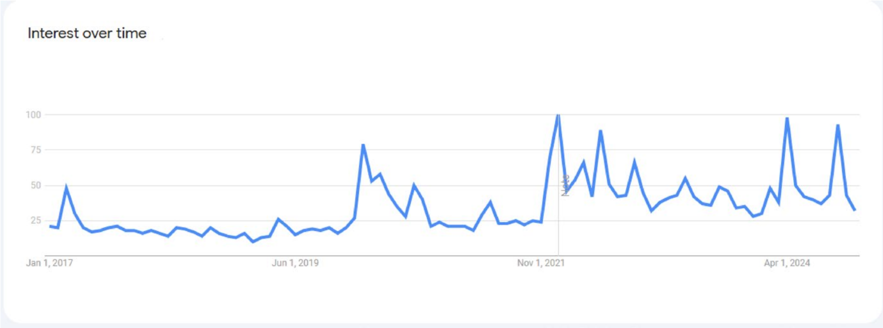


Fig. 11 Google trends interest over time (2017–2024) for hate speech-related queries. *Note*: This line chart illustrates public interest in hate speech-related search terms based on Google Trends data from January 2017 to 2024. The spikes represent periods of heightened search activity, often coinciding with real-world events, controversies, or online incidents that triggered public discourse on hate speech. Notable increases are observed in late 2018, mid-2020, and throughout 2021–2023, suggesting growing awareness and concern around online abuse, particularly in multilingual and social media contexts

standing. These techniques, however, confront challenges such as data dependency, bias, and the necessity for ongoing adaptation.

Multimodal approaches [37, 38] combine textual information with other data forms like images and audio to improve hate speech detection in multimedia content. While this improves coverage, it introduces new complexities such as alignment of different data types, higher training costs, and interpretability issues. Additionally, storing and processing multimedia data raises privacy and scalability concerns. Contextual analysis researchers, including [39, 40], focus on understanding the intent behind a message by incorporating conversational, social, and temporal context. These methods are more accurate but come with increased computational cost and privacy risks, especially when user histories or thread-level context are involved. They are also sensitive to cultural differences, irony, and subjectivity.

Furthermore, scholars such as [41] and [42] have worked on adapting detection techniques across multiple languages, addressing challenges like linguistic complexity and cultural sensitivity in cross-lingual hate speech detection. It is significant for diverse countries like India, where multiple regional languages coexist. However, these models often face challenges due to limited training data in low-resource languages, bias from dominant languages, and the difficulty of capturing cultural nuances through translation. Ethical issues also arise when dealing with politically or socially sensitive language.

Each approach has its own set of advantages and limitations, and the choice of which method to use depends on factors such as the nature of the content, available resources, the desired level of accuracy, and the specific use case. Combining multiple approaches or developing hybrid systems is a common strategy to address the limitations of individual methods and enhance overall performance. Additionally, ensuring fairness and ethical considerations in hate speech detection remains a critical concern, especially given the potential biases inherent in some techniques. Overall, most existing approaches face challenges in generalization, fairness, interpretability, and scalability, especially when applied in multilingual, real-time, and socially complex

Table 8 Descriptions of various hate speech detection methods

Authors	Methods	Description	Techniques	Limitations	Practical deployment limitations
Corazza et al. [45], Recupero et al. [10], Vigna1 et al. [34], Hayaty et al. [35], Qureshi et al. [95], Pronoza et al. [96], Kapil et al. [97], Corazza et al. [98]	Lexical and Text-Based Approaches	Lexical and text-based methods for finding hate speech look closely at the words, short phrases, or patterns in the text to spot hate speech. These methods focus on the words and how they are put together without thinking much about the bigger picture or the whole message	Profanity and Offensive Word Lists, N-grams and Text Patterns, Regular Expressions, Sentiment Analysis	Limited context, False Positives, Inability to adapt o new or unexpected forms	Often fails in social media and meme contexts where hate is implicit, ironic, or slang-driven. Cannot adapt to evolving language or domain shifts Not scalable for multilingual settings
Kumaresan et al. [4], Antonakaki et al. [99], Kohli et al. [43], Ahammed et al. [18], Yadav et al. [100], Arcila-Calderón et al. [21], Istaiteh et al. [75], Hasan et al. [36], Pannerselvam et al. [101], Ayo et al. [102], Indurthi et al. [103], Alsharef et al. [45], Fan et al. [104], d'Sa et al. [105]	Machine Learning and Natural Language Processing (NLP) Techniques	Machine Learning enables computers to learn from data, while NLP focuses on language understanding and interaction	Supervised Learning, Unsupervised Learning, Neural Networks, Word Embeddings	Data Dependency, Bias, Lack of Common Sense, Context Understanding, Resource-Intensive, Interpretability, Data Privacy, Multilingual Challenges, Constant Adaptation, Overfitting	Deployment faces bias propagation, especially in under-represented communities Multilingual performance is limited by resource inequality

Table 8 (continued)

Authors	Methods	Description	Techniques	Limitations	Practical deployment limitations
Chhabra et al. [106], Guellil et al. [37], Gambäck et al. [38], Suryawanshi et al. [107], Jing et al. [108], Rahate et al. [109], Gandhi et al. [110], Hamza et al. [111], Sai et al. [112], Dwivedy et al. [113], Rana et l.[114], Prasad et al. [115], Chhabra et al. [106]	Multimodal Approaches	Combining textual information with other modalities, such as images and audio, to detect hate speech in multimedia content	Feature Fusion, Multimodal Preprocessing, Deep Learning Architectures, Attention Mechanisms, Transfer Learning, Cross-modal Learning, Multimodal Interpretability	Data Complexity, Data Availability, Training Complexity, Intermodality Relationships, Model Overhead, Data Quality and Consistency, Domain-specific Challenges, Limited Interpretability, Privacy Concerns, Scalability, Generalization, Resource Constraints	High computational overhead and platform-specific tuning are required Data annotation is expensive across modalities. Multimodal models lack generalizability and are difficult to scale for real-time moderation across platforms
Saleh et al. [116], Alshalan et al. [117], Calderón et al. [118], Masud et al. [119], Waniarachchi et al. [120], Ghozali et al. [121], Sahin et al. [122], Khullar et al. [123], Valle-Cano et al. [124], Arco et al. [125], Marrugo-Tobón et al. [126], Wiedlitzka et al. [127], Okulska et al. [128], Saleh et al. [116]	Contextual Analysis	Analyzing the context in which content is posted to understand the intent behind the language and differentiate between hate speech and legitimate discussions	Semantic analysis, Entity Recognition, Coreference Resolution, Dependency Parsing, Topic Modeling, Named Entity Recognition (NER), Coreference Chains, Temporal analysis, Pragmatics and Speech Acts, Discourse Analysis, Contextual Word Embeddings, Transformer-based models	Ambiguity, Complexity, Cultural and Contextual Variations, Irony and Sarcasm, Limited Common Sense, Generalization Challenges, Subjectivity, Privacy Concerns, Real-time Constraints, Multimodal Integration, Bias and fairness, Context Shifts, Data Availability	Real-time implementation is costly due to complexity Models often misinterpret sarcasm or nuanced hate in different cultural contexts Performance degrades in informal or multilingual discourse

Table 8 (continued)

Authors	Methods	Description	Techniques	Limitations	Practical deployment limitations
Basile et al. [129], Guellil et al. [37], Rodríguez et al. [39], Saha et al. [40], Bestgen et al. [41], Awal et al. [42], Renjit et al. [130], Arango et al. [131], Corazza et al. [98]	Cross-lingual Hate Speech Detection:	Adapting detection techniques to multiple languages and addressing the challenges of linguistic diversity	Transfer Learning, Language-Agnostic Features, Multilingual Embeddings, Cross-lingual Knowledge Transfer, Rule-Based Approaches, Active Learning, Parallel Corpora	Data Scarcity, Bias and Representativeness, Language Complexity, Cultural Sensitivity, Imbalanced Languages, Multimodal Challenges, Limited Generalization, Resource Intensity, Continual Adaptation, Ethical Consideration	Low-resource languages lack labeled datasets. Models trained in high-resource languages do not generalize well Deployment in multilingual nations requires domain-specific adaptation and ethical alignment

environments like India. Continuous adaptation, context-awareness, and culturally sensitive design remain essential for building robust hate speech detection systems.

6 Datasets for hate speech detection

Hate speech detection in India and multilingual contexts heavily relies on diverse datasets representing different languages, platforms, and annotation schemes. Table 9 summarizes the significant datasets used in recent research.

The datasets across multiple social media platforms, including YouTube, Facebook, and Twitter, cover monolingual and code-mixed language settings. These datasets vary in size, annotation types (binary or multiclass classification), and language combinations, providing valuable resources for benchmarking hate speech detection techniques. Monolingual datasets focus on individual languages such as Bengali, Tamil, and Malayalam, enabling detailed studies specific to each language's nuances in hate speech detection. Code-mixed datasets, on the other hand, incorporate combinations like Hindi-English, Tamil-English, and Kannada-English, reflecting the inherently multilingual nature of social media communication in India. Multilingual datasets go beyond Indian languages to include global languages such as English, Spanish, and Bulgarian, facilitating comparative studies and enabling cross-lingual research in hate speech detection.

The significant issue of the dataset is the imbalance in class distributions, as many datasets contain fewer hate speech samples compared to non-hate speech, which can lead to biased models. Additionally, some datasets incorporate multimodal information, such as textual and visual content, further complicating preprocessing and modeling efforts by requiring techniques that can effectively handle diverse data types. These datasets have played a crucial role in advancing hate speech detection by offer-

Table 9 Descriptions of datasets in hate speech detection

Author	Dataset	Size	Class	Language	Strenght	Weaknesses
Romim et al. [132]	Facebook, YouTube	30,000 comments	Binary (Hate or Non-hate)	Bengali	Provides substantial Bengali content, a low-resource language often excluded from hate speech corpora; suitable for monolingual benchmarking	Binary labels oversimplify complex hate speech nuances; they lack contextual diversity across platforms or discourse styles
Bohra et al. [32]	Twitter	4,575 tweets	Binary (Hate or Non hate)	Hindi- English (Code-mixed)	Captures authentic code-mixed social media language reflective of Indian urban discourse; useful for testing multilingual transfer learning	Dataset size is insufficient for deep learning models; it lacks fine-grained categorization and linguistic annotations
Hande et al. [133]	KanCMD (Youtube)	7,671 comments	Multiclass	Code-mixed (Kannada and English)	Includes rich code-mixed data and multiple hate categories, supporting nuanced classification in a regional language	Limited domain scope (YouTube); relatively low size hampers generalizability and cross-domain training
Chakravarthi et al. [134]	Youtube	15,744 comments	Multiclass	Code-mixed (Tamil and English)	High-quality annotations for hate subcategories; designed for shared tasks, thus offering strong benchmarking potential	Class imbalance and annotation subjectivity may introduce noise; they lack metadata on the annotator's background or context
Chakravarthi et al. [135]	DravidianLang-Tech Data (Youtube)	Tamil 43,919, Malayalam 20,010, and Kannada 7,772 comments	Multiclass	Multilingual (Malayalam, Tamil, and Kannada)	Offers robust multilingual coverage of South Indian languages, ideal for cross-lingual and regional model evaluation	Data is platform-specific (YouTube), potentially limiting generalizability; skew in class distribution affects fairness

Table 9 (continued)

Author	Dataset	Size	Class	Language	Strenght	Weaknesses
Chakravarthi et al. [136]	Youtube	English 28,451, Tamil 20,198, Malayalam 10,705	Multiclass	Multilingual (Malayalam, Tamil, and English	Enables comparative analysis between Indian languages and English; supports multilingual model alignment	Annotation consistency and clarity are not well-documented; they lack entity-level or context-aware metadata
Thomas Mandla et al. [137]	HASOC (Youtube and Twitter)	2963	Binary class	Tamil and Malayalam	Part of a competitive shared task; annotations follow strict guidelines ensuring quality control	Dataset size is too small for training; lacks intra-class variability and depth in label structure
Kumaresan et al. [138]	YouTube and Twitter	28,424 English, 17,715 Tamil, 9,918 Malayalam, 6,176 Kannada and 1,650 in Spanish comments	Binary class	English, Tamil, Malayalam, and Kannada	Broad multilingual and cross-platform design (Twitter + YouTube) enhances robustness in real-world model testing	Label granularity is limited to binary; the absence of socio-linguistic context limits semantic understanding

ing diverse linguistic and cultural perspectives, enabling more inclusive and comprehensive research. However, there is significant scope for future improvements. Expanding datasets to include underrepresented languages can help address gaps and ensure the broader applicability of detection models. Moreover, they ensure compliance with ethical and regulatory guidelines, respecting privacy and consent. High-quality datasets, which are well-annotated, diverse, balanced, and representative of real-world scenarios, are essential for training and evaluating hate speech detection models. These datasets include accurate labels, address different types of hate speech across various contexts, and minimize biases, ensuring fairness and inclusivity in model training.

Figure 12 shows the significance of high-quality datasets in addressing the challenges of hate speech in the digital landscape. Such datasets are the foundation for training, evaluating, and refining hate speech detection models, ensuring they are accurate, fair, and capable of tackling the diverse hate speech across different platforms. Ensuring data trustworthiness by sourcing data ethically, labeling it with care, and minimizing bias is a basis for building high-quality datasets. This commitment not only enhances model reliability but also ensures that the models effectively address real-world challenges. High-quality datasets play a vital role in every stage of the hate speech detection pipeline. They provide reliable and representative examples for model training, enabling systems to distinguish between hateful and non-hateful content effectively. The quality of training data directly influences model performance, improving precision and recall.

Fig. 12 Importance of a high-quality dataset. *Note*: Fig. 12 illustrates the critical role of high-quality datasets in hate speech detection. These datasets form the backbone for training, evaluating, and refining models, ensuring accuracy, fairness, and contextual relevance across diverse platforms and languages. Ethical data sourcing, careful annotation, and bias mitigation are essential to enhance trustworthiness and model performance

Additionally, diverse datasets help mitigate biases by incorporating multiple perspectives, thereby promoting fairness in model predictions. They also support better generalization, allowing models to handle new, unseen data across different languages, dialects, and platforms. Evaluation processes depend on trustworthy datasets for objective benchmarking of models. Moreover, ethical and regulatory compliance in data collection, as well as respecting privacy, consent, and legal standards, is essential for responsible AI development. As hate speech varies across contexts, high-quality datasets ensure real-world applicability, enabling models to perform effectively in dynamic online environments. The development and curation of these datasets are ongoing efforts to advance content moderation and promote online safety [43, 44].

Table 9 illustrates the datasets included in this review on hate speech in Indian languages, mainly using content from social media platforms like Twitter and YouTube. While some datasets use simple binary labels such as “hate” or “not hate,” others offer more detailed classifications. Although multiclass labeling provides deeper insight into different types of hate speech, it is harder to annotate accurately and can lead to inconsistencies. Many datasets include code-mixed content, such as Hindi-English or Tamil-English, which better reflects real-world language use online. However, several Indian languages remain underrepresented, making it difficult to develop models that work well across all linguistic communities. The size of these datasets also varies widely. Although larger datasets are better for training, they can suffer from poor annotation quality, especially when volunteers do labeling without strict guidelines. In addition, most datasets do not adequately cover culturally sensitive forms of hate speech, such as caste-based or gender-based abuse, and they often miss newer terms or evolving patterns of online hate. Since most datasets focus on specific platforms like Twitter or YouTube, models trained on them may not perform well on other social media sites. These limitations highlight the need for more inclusive, balanced, and regularly updated datasets that consider linguistic, cultural, and platform-specific diversity to build effective and fair hate speech detection systems.

7 Challenges and issues

Hate speech detection faces several challenges, including ensuring fairness, privacy, and freedom of expression while avoiding overblocking. Overblocking refers to removing or blocking legitimate content that is not actually hateful, often due to false positives, lack of cultural or contextual understanding, or overly strict filtering. For instance, harmless jokes, cultural expressions, or political criticism may be mistakenly flagged as hate speech, which suppresses free expression and disproportionately affects marginalized communities. It must also address multilingual barriers, context sensitivity, and the impact on marginalized communities. Furthermore, issues such as false positives and negatives, legal and ethical frameworks, and the need for transparency and accountability make content moderation a highly complex task. It may accidentally censor politically or socially sensitive content instead of actual hateful material. It is challenging to address bias and ensure fairness in detection models because these systems often pick up biases from the data they are trained on [45]. Detecting harmful intent in varying contexts adds another complexity, as automated systems must distinguish between hate speech and counter-speech. Adapting hate speech detection to encompass multiple languages, cultures, and legal frameworks presents additional challenges. One of the primary concerns is bias and fairness, as models often inherit and propagate societal biases present in training data, which can disproportionately affect marginalized communities. The multilingual nature of Indian languages with variations in script, grammar, and context adds significant complexity to the development of robust detection systems. Context sensitivity is another critical factor, as many hateful expressions rely on implicit, coded, or culturally nuanced language that machines struggle to interpret accurately. The issue of false positives and false negatives further complicates the matter, overly strict models may misclassify non-hateful content, while lenient ones may fail to detect harmful speech.

Balancing freedom of expression with content moderation is difficult, as too much censorship can stop people from sharing honest opinions. At the same time, blocking or removing content without clear reasons can reduce trust and make open communication harder. Legal and ethical concerns also arise, as the lack of uniform legal frameworks and culturally sensitive policies leads to inconsistent enforcement. Transparency and accountability in moderation practices are essential to ensure fairness and build public confidence. Additionally, privacy concerns emerge from the need to analyse user-generated content while protecting personal data. Community and user feedback mechanisms are vital for participatory moderation, but their integration into automated systems is still limited. Lastly, the phenomenon of praises for avoidance, where individuals use mild or coded language to avoid detection by hate speech systems, presents a significant challenge. Together, these challenges highlight the urgent need for interdisciplinary, ethically grounded, and context-aware approaches to hate speech detection. The ethical implications of automated content removal, such as false positives and negatives, highlight the need for a balanced approach to mitigate [46] hate speech effectively. This requires a comprehensive strategy integrating technological innovations, human moderation, legal frameworks, user education, and active community participation. Ensuring the ethical development and deployment of hate speech detection systems demands adherence to fundamental principles such as freedom of expression, privacy, and fairness [47].

The biased and debiased methodologies, descriptions, and metrics used to evaluate bias are highlighted in Table 10. Bias and fairness in algorithms pose significant challenges in developing and using machine learning and artificial intelligence systems. These concerns often lie in biased training data, as algorithms can inadvertently learn and perpetuate existing biases [48]. Biased algorithms have the potential to discriminate, leading to unfair treatment, unequal opportunities, and the reinforcement of stereotypes. Defining fairness in algorithms is complex, as different fairness definitions may conflict [49]. Protected attributes like race or gender should never be used for discrimination, but identifying and handling these attributes can be complicated.

This is because such information is often implicit, context-dependent, or sensitive, making it difficult to detect accurately without risking privacy or misclassification. Transparency and accountability become challenging because machine learning models often work like “black boxes,” making it hard to understand how they make decisions. Additionally, overfitting can make the problem worse by reinforcing existing biases in the data [50]. Feedback loops can amplify bias over time, and a trade-off between fairness and accuracy complicates mitigation efforts. Addressing bias and fairness requires diverse and representative training data, fairness-aware machine-learning techniques, ethical considerations, and ongoing audits of algorithms. Ensuring fairness in algorithmic systems is not just a technical task, but also an important ethical concern, especially as data-driven technologies play a vital role in decision-making [51]. Automated content moderation systems can lead to various unintended consequences.

One significant consequence is overclocking, where these systems censor legitimate speech, impinging upon freedom of expression. Conversely, they can unblock

Table 10 Various biases and debiased methods

Bias	Description	Bias evaluation method	Debiased method
Human-produced texts	Human-produced texts in large-scale language corpora are the origins of implicit associations and biases. Natural language processing models, which train language representations through Unsupervised Learning, uncover indirect patterns in data and grasp implicit associations [139]	Word Embedding Association Test, Valence Assessing Semantics Test Implicit Association Test [139], Error rate equality differences, Curve or AUC, Pinned AUC, Pinned AUC Equality Difference [140]	Name-based Counterfactual Data Substitution [141], Debiased word embeddings [142], Counterfactual Data Augmentation, Ensemble-based debiasing [143], LEARNED-MIXIN [144], AFLITE [145], Multi-task learning [146]
Gender-biased	Gender bias refers to the fixed prejudice towards a specific gender, manifested through the attitudes and behaviors of individuals and the broader norms and practices of societal systems [141]		
Adult language	It negatively impacts society, especially when treating diverse individuals differently [147]		
Racial Bias	Racial bias is prejudiced attitudes or behaviors towards individuals based on race, leading to discrimination and systemic inequalities [148]		
Sampling bias	Sampling bias occurs when a sample does not represent the whole population, leading to skewed data and potentially misleading conclusions about broader trends or characteristics [149, 150]		
Algorithmic bias	The design and implementation of the algorithms can prioritize specific attributes, potentially leading to biased and unfair outcomes [48, 49]		
Lexical Bias	Lexical bias is a tendency in language where specific word choices or phrasings subtly influence perception and communication, often reflecting underlying attitudes or cultural contexts [150, 151]		
Social bias	Social bias is the unjust treatment of certain people based on perceived group membership or characteristics [139]		

This table compiles various bias types, their descriptions, commonly used evaluation metrics, and corresponding debiasing techniques referenced in recent literature.

harmful content, permitting hate speech, harassment, or misinformation to persist [52, 53]. A holistic approach is needed to address these issues, combining automated systems with human moderation, transparency in content policies, transparent appeal processes, and ongoing ethical considerations to create a balanced and safe online environment [54].

8 Legal and regulation in India

In India, managing hate speech on social media presents a complicated challenge for the legal system. The country's regulatory strategy merges laws from the colonial era with specific regulations for online communications. This approach aligns with the judicious restrictions outlined in the Indian Constitution and tends to mirror European legal models rather than the more liberal framework prevalent in the United States [55]. The nation employs various legal instruments, notably the Indian Penal Code (IPC) of 1860 and the Information Technology (Intermediary Guidelines and

Digital Media Ethics Code) Rules of 2021. Sections 153A, 295A, 298, and 505 of the IPC penalize speech that provokes violence or fosters hatred on grounds such as religion, race, and language, threatening communal harmony. The 2021 IT Rules require intermediaries to promptly remove or restrict access to problematic content within 36 h of a complaint, under threat of losing their legal immunity [56]. The identification of hate speech now leans more on algorithmic and keyword searches than on classical legal policies influenced by the policies of global digital platforms. This shift is evident in historical and recent legal developments, such as the Indian Cinematograph Act of 1918 and the significant 2015 Supreme Court ruling in the Shreya Singhal case. Law enforcement and intelligence agencies conduct selective internet monitoring, heavily relying on the Electronic Media Monitoring Centre (EMMC) of the Union Ministry of Information and Broadcasting to observe social media trends. This approach is supplemented by a mandate for intermediaries to actively prevent the spread of inflammatory and hateful content, as required by the Information Technology Act of 2015 [55].

9 Future directions

The field of hate speech detection offers a rich landscape for further research and development. One prominent avenue is multimodal detection, which involves enhancing the capability to identify hate speech in various forms, including text, images, audio, and video [57]. This expansion to different modalities requires developing models that effectively integrate information from these diverse sources, catering to the evolving ways hate speech is disseminated online [58]. Additionally, cross-lingual and multilingual hate speech detection is an area of continued growth, where researchers aim to improve the adaptability of models to the linguistic diversity of online content. This involves addressing language-specific nuances and cultural variations to create more effective detection systems globally [4, 59]. Exploring the explainability and interpretability is another crucial dimension. As automated hate speech detection systems play an increasingly central role in content moderation, it becomes imperative for users to understand and trust the decisions made by these systems. Developing models that provide transparent explanations for their judgments is critical to creating trust and accountability. Fairness and bias mitigation remain at the forefront of research, ensuring that hate speech detection models do not target specific demographic groups disproportionately. Ethical considerations are an overarching theme, impacting various aspects of development, from safeguarding user privacy to understanding the interplay between combating hate speech and protecting freedom of expression. In the evolving landscape of online hate speech, there is also a need for robust, adaptable models that can withstand tactics employed by those committing hate speech. The dynamics of hate speech continue to change, requiring dynamic data collection strategies and continual learning approaches to keep detection systems practical and up-to-date. Collaboration with users, community moderators, and ethical experts is essential to align technology development with user needs and ethical standards. These areas for further research reflect the continuous evolution of fields and the growing emphasis on creating responsible, effective, and user-centered hate speech detection solutions.

10 Implications and recommendations

Hate speech detection has significantly broader implications for society and social media platforms, touching on various aspects of online communication, community well-being, and freedom of expression. Some critical implications of hate speech detection on society and social media platforms include several key areas: Effective hate speech detection systems are crucial for mitigating harm caused by online hate speech, including harassment, discrimination, and violent incitement. By doing so, these systems contribute to creating safer and more inclusive online environments. They play a particularly vital role in protecting vulnerable communities that are disproportionately affected by hate speech by identifying and removing harmful content. These detection systems also raise critical questions about the balance between freedom of expression and content moderation, a complex and ongoing challenge for society and social media platforms. This balance is further complicated by the need for transparency and accountability in content moderation, as platforms are increasingly pressured to explain their decision-making processes and provide recourse for users. Moreover, to ensure fairness, hate speech detection systems must avoid algorithmic biases that can disproportionately impact specific groups. This fairness is crucial for preventing the removal of non-discriminatory content. The impact of these systems on user behavior is also significant. Users may self-censor or refrain from engaging in discussions due to the fear of automated content removal, potentially limiting open discourse. The real-world impact of online hate speech is undeniable, with the potential to incite violence and radicalization. Therefore, detection systems that can identify and prevent the spread of harmful content are vital. However, the ethical implications of hate speech detection are multifaceted, spanning privacy, consent, and freedom of expression issues. These ethical considerations are becoming more prominent in discussions about content moderation.

The legal frameworks and regulations surrounding hate speech detection intersect with the effectiveness and ethics of these systems. Countries and regions are crafting laws that shape how platforms moderate content, impacting the development and implementation of detection systems. Developing effective hate speech detection systems also necessitates interdisciplinary collaboration among experts in natural language processing, ethics, sociology, and law. This collaborative approach ensures that multiple perspectives are considered when creating these systems [60]. Lastly, the user experience is significantly affected by the accuracy and fairness of detection systems. Inaccuracies or biases in these systems can lead to frustration and distrust, affecting how users perceive and engage with online platforms. Therefore, these systems must be carefully designed and continuously improved to maintain user trust and ensure a positive online experience.

A knowledge graph in Fig. 13 centered on Hate Speech Detection, which branches into two main categories: Societal Implications and Platform Implications. Under Societal Implications, the graph highlights the goal of Mitigating Harm, which is

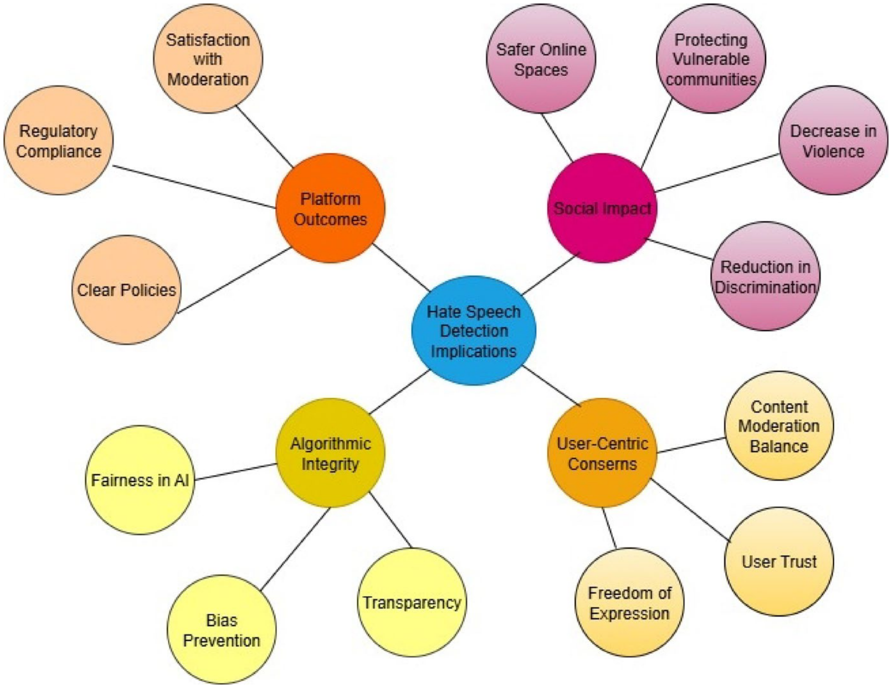


Fig. 13 Knowledge graph for hate speech detection implications. *Note*: A knowledge graph outlining the broader implications of hate speech detection systems. It maps key thematic areas such as social impact, user-centric concerns, platform outcomes, and algorithmic integrity. The graph highlights interrelated concepts like fairness, bias prevention, user trust, freedom of expression, and regulatory compliance, emphasizing the multifaceted consequences of automated moderation systems on digital platforms and society

further linked to reducing discrimination and promoting safer online environments. Additionally, it emphasizes the role of hate speech detection in Protecting Vulnerable Communities through Enhanced Safety Measures. It outlines the Real-World Impact as leading to Decreased Violence and fostering Positive Social Change. On the platform side, the graph delineates the complexities of Freedom of Expression, including the challenges of Content Moderation Balance and Regulatory Compliance. It underscores the need for transparency in platform operations, leading to clear policies and fostering user trust. The importance of addressing Algorithmic Bias is also captured, with connections to Fairness in AI and Bias Prevention strategies. Finally, the graph considers user experience, where the accuracy and fairness of detection systems affect satisfaction with Moderate and overall engagement quality on platforms. This structured visualization of relationships underscores the multifaceted impacts of hate speech detection on society and online platforms.

The implications of hate speech detection extend far beyond the realm of technology. For society, these detection systems represent a critical tool in the fight against online hate and discrimination, promoting a safer and more inclusive digital environment. They can potentially protect vulnerable individuals and marginalized communities, making online spaces more open to discourse. However, implementing these

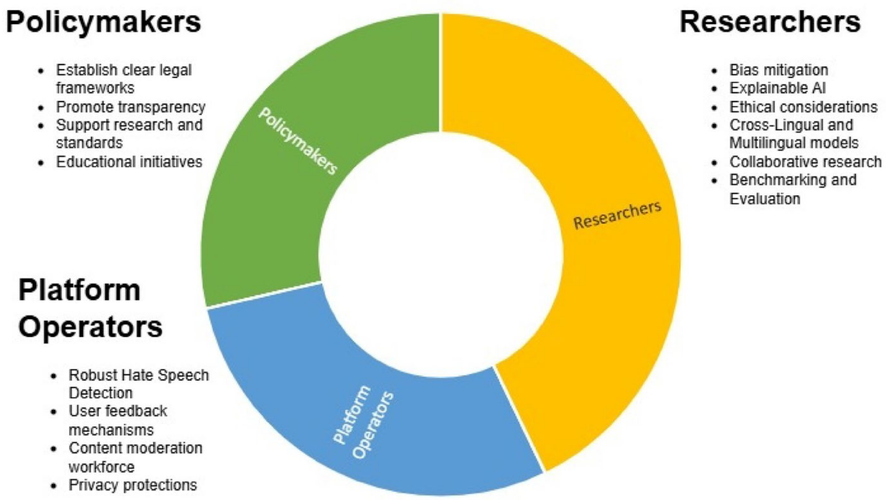


Fig. 14 Recommendation for addressing hate speech on social media

systems also raises concerns about free speech, censorship, and privacy, highlighting the delicate balance between combating hate speech and safeguarding fundamental rights. For social media platforms, hate speech detection is integral to maintaining their reputations and user trust. The effectiveness of these systems influences their ability to create spaces where users feel safe and respected, impacting user engagement and the success of their platforms. Striking the right balance between moderation and user autonomy is essential for the long-term sustainability of these platforms and ethical operation in an increasingly interconnected world.

The recommendations for policymakers, platform operators, and researchers involved in hate speech detection are illustrated in Fig. 14. To effectively combat hate speech, policymakers must establish clear and up-to-date legal frameworks that define hate speech parameters while safeguarding freedom of expression. This approach should be complemented by promoting transparency in online platforms, mandating them to disclose their hate speech detection efforts, including the number of removed posts, appeal processes, and overall system performance. Additionally, policymakers should allocate resources for research into hate speech detection and collaborate with experts to set industry-wide standards. Educational initiatives promoting digital literacy and responsible online behavior are also crucial.

Platform operators should focus on developing robust hate speech detection systems that are not only advanced, ethical, culturally sensitive, and multilingual. These systems must incorporate user feedback, allowing for continual improvement. The mental health and training of the content moderation workforce should be a priority, ensuring they are well-equipped to handle the challenges of their roles. Privacy protections are also vital, with the implementation of technologies that safeguard user data while facilitating effective hate speech detection.

Researchers play a vital role in mitigating biases in detection models, striving for fairness and equity. The development of explainable AI is crucial to fostering user trust and understanding of automated content moderation decisions. Ethical considerations must be at the forefront of hate speech detection strategies, particularly the impact on marginalized communities, privacy, and broader societal context. Developing cross-lingual and multilingual models is essential to address the diverse linguistic landscape of online communities. Finally, researchers should engage in collaborative, interdisciplinary research and improve benchmark datasets and evaluation metrics, ensuring that hate speech detection models meet high standards of efficiency and effectiveness. Together, these recommendations can contribute to developing effective, ethical, and user-centered hate speech detection systems that balance promoting online safety and preserving fundamental rights and freedoms.

11 Conclusion and future work

This Systematic Literature Review (SLR) gives a broad overview of the growing field of hate speech detection and highlights the important role of technology in tackling this widespread issue. As the online world continues to expand, key insights show that multilingual and multimodal approaches are becoming more important, especially for Indian languages and various types of content like text, images, audio, and video. The review points out the need for strong technologies to detect hate speech and also emphasizes fairness, transparency, and privacy in detection systems to ensure ethical outcomes. Recent research is increasingly focused on reducing bias and making AI decisions more straightforward to understand through Explainable AI, which provides clear reasons for the system's actions. The field is also moving towards including user-centered approaches, community feedback, and cross-lingual models. Moving forward, it is important to balance the need for freedom of expression with the need to combat hate speech, and this requires clear legal rules and educating users. While hate speech detection systems are key to making online spaces safer, we must also address the broader social issues they raise, such as privacy, censorship, and user freedom, through teamwork among policymakers, platform operators, and researchers.

In conclusion, effective hate speech detection on social media is crucial for creating a safe and inclusive online environment, especially for marginalized groups who are often targeted. By quickly identifying and removing harmful content, these systems can reduce the spread of harmful and discriminatory messages, leading to a more respectful online space. They also help build trust in social media platforms, which is important for their long-term success. However, these systems must carefully balance free speech, privacy, and safety. Achieving this balance requires collaboration from all involved parties. The challenge is significant, but a shared responsibility is vital for creating a digital space where freedom of expression is respected and all users are treated fairly.

Future research in hate speech detection can explore several key areas. First, there is a need for more diverse datasets that include underrepresented languages and cultures, ensuring broader applicability across different communities. Further advancements in multimodal detection systems can integrate text, audio, and visual data to improve contextual understanding and detection accuracy. Moreover, addressing the challenge of balancing privacy with efficient detection will require developing privacy-preserving technologies, such as federated learning. More research is needed to refine Explainable AI techniques to ensure that models are effective and transparent. Finally, a continued focus on interdisciplinary collaboration involving legal experts, social scientists, and technologists will be crucial to creating frameworks that respect freedom of expression while combating harmful content online.

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Declarations

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